Session 9 — Non-Current Assets I: Property, Plant and Equipment, Recognition and Initial Cost (IAS16)

By the end of this session you must know IAS16's objective and scope, the definition and recognition criteria for property, plant and equipment (PP&E), which elements make up an asset's initial cost and which costs must be excluded, the component approach to depreciation, and how borrowing costs on qualifying assets are capitalised under IAS23.

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1. Objective and Scope of IAS16

Objective of IAS16 (par 1): to prescribe the accounting treatment for property, plant and equipment, so that users of financial statements can discern information about an entity's investment in its PP&E. The principal issues are the recognition of PP&E, the determination of their carrying amounts, and the determination of depreciation charges and impairment losses.

IAS16 applies to property, plant and equipment generally, but it explicitly does not apply to:

- PP&E classified as held for sale (IFRS5).
- Biological assets related to agricultural activity (IAS41).
- The recognition and measurement of exploration and evaluation assets (IFRS6).
- Mineral rights and mineral reserves such as oil, natural gas and similar non-regenerative resources.
- Investment property (IAS40, covered separately in Session 10).

2. Definition and Recognition of PP&E

Property, plant and equipment (IAS16 par 6): tangible assets that (a) are held by an enterprise for use in the production or supply of goods or services, for rental to others, or for administrative purposes; and (b) are expected to be used during more than one period.

- The cost of an item of PP&E is recognised as an asset if and only if **both** of the following hold: the cost of the item can be measured reliably, and it is probable that future economic benefits associated with the item will flow to the entity.

Example: a government mandates that a company install pollution-reduction equipment costing €350,000. Applying the two recognition criteria, the cost is reliably measurable, and it meets the future-economic-benefits criterion because the expenditure is incurred to obtain (or maintain) the economic benefits from the entity's other assets — the plant could not otherwise legally keep operating. The cost of the pollution-reduction equipment is therefore capitalised as PP&E, not expensed.

- IAS16 restates the same two conditions at the recognition stage (par 7): probable future economic benefits associated with the asset, and a cost (or fair value, if using the revaluation model) that can be measured reliably.

3. Elements of Initial Cost

Three elements of cost (IAS16 par 15-20): purchase price; directly attributable costs; and the initial estimate of the costs of dismantling and removing the item or restoring the site on which it is located.

- Directly attributable costs are those necessary to put the asset into operation for normal use as intended by management: professional fees of architects and engineers, delivery and handling costs, site preparation costs, and installation/calibration costs.
- Costs that are not directly attributable cannot be capitalised — administrative costs and overheads are excluded, as are costs not directly necessary for construction such as feasibility studies, staff training costs, and costs relating to abnormal waste of material and labour in self-constructed assets.
- The estimated cost of dismantling, removing and restoring the site is recognised as a provision on the liability side, and is amortised over the asset's life as part of its cost.

Worked example (machine cost): a company buys a machine for €100, plus 7% sales tax, €10 transport to the plant, €5 storage (plant not ready — excluded, see Section 4), €3 labour and €2 materials to calibrate the machine, and €11 consultancy fees related to the choice of machine and its calibration. The capitalised cost is €133: invoice plus tax (100 + 7 = 107), plus transportation (10), plus calibration (3 + 2 = 5, appears in the source total as “Calibration 5”), plus professional fees (11) — totalling 107 + 10 + 5 + 11 = 133. Storage costs of €5 incurred because the plant was not yet ready are excluded as not directly attributable to bringing the asset to its intended condition.

Worked example (equipment purchase, 31 July): a company buys equipment with a manufacturer’s price of €500,000, less a trade discount of €30,000 (net €470,000), plus a delivery charge of €4,300, installation and testing charges of €24,500, and small spare parts of €5,200 — total capitalised cost €504,000. A separate €36,000 servicing contract fee for the year to the following 31 July is **not** part of the asset’s cost: it is a period expense for a future service, not a cost of bringing the asset into use.

4. Costs Excluded From Initial Recognition

Recognition of costs in the carrying amount of an asset ceases when the asset is capable of operating as intended by management — not necessarily when it is actually in full use.

Three categories of cost cannot be included in initial recognition once that point is reached:

- Costs incurred for an asset that is not yet in use, or not yet operating at full capacity, but is already capable of operating in the manner intended by management.
- Initial operating losses, such as those incurred while demand for the item’s output builds up.
- The costs of relocating or reorganising part or all of the entity’s business.

Example (supermarket renovation): a company acquires a new supermarket that needs renovation, closing the store for three months. Renovation costs (necessary to bring the store into the condition intended by management) can be capitalised. But the wages of staff preparing the shop before opening, and utility costs incurred before opening, cannot be capitalised — they are operating costs that would have been incurred in any case once the shop opened, not costs necessary to bring the store into an operational condition.

5. Component Approach to Depreciation

Component approach (IAS16 par 43): each part of an item of PP&E with a cost that is significant in relation to the total cost of the item must be depreciated separately.

Example: for an aircraft, the engine (nacelle) typically has a different useful life than the seats or the airframe, so each significant component is depreciated on its own schedule rather than the asset being depreciated as a single undifferentiated block.

6. Capitalising Borrowing Costs (IAS23)

Borrowing costs (IAS23): interest and other costs that an entity incurs in connection with the borrowing of funds.

Qualifying asset: an asset that necessarily takes a substantial period of time to get ready for its intended use or sale.

- Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset must be capitalised as part of that asset’s cost; all other borrowing costs are expensed in the period incurred.
- The capitalisation rate depends on the funding source:
  - For specific borrowings taken out directly to fund the asset, the borrowing cost is simply the interest expense on those specific borrowings.
  - For general borrowings that partly fund the asset, the borrowing cost is computed using the weighted average interest rate on general borrowings during the capitalisation period.

Capitalised borrowing cost (general borrowings) = Σ (Qualifying expenditure × Weighted average borrowing rate × Months outstanding ÷ 12)

where: - the weighted average rate is computed across all general borrowings outstanding during the period, weighted by each borrowing’s principal amount; - each tranche of qualifying expenditure is capitalised only from the date it was incurred to the end of the capitalisation period, hence the months-outstanding fraction. - reading it as a calculation: for each tranche of qualifying expenditure, multiply it by the weighted average borrowing rate, then scale that down to only the months it was actually outstanding during the period; sum the tranches to get the total capitalised borrowing cost.

Worked example: a company has three general borrowings outstanding through the year to 30 June: €800,000 at 7.5%, €500,000 at 9%, and €1,200,000 at 8.5%. Total borrowing cost = (800,000 × 7.5%) + (500,000 × 9%) + (1,200,000 × 8.5%) = €207,000, on total borrowings of €2,500,000, giving a weighted average rate of 8.28%. Construction of a qualifying asset began 1 October with an expenditure of €300,000, and a further €240,000 was spent on 1 February; both were financed from general borrowings, and construction was still underway at year-end (30 June). Capitalised borrowing cost = (300,000 × 8.28% × 9/12) + (240,000 × 8.28% × 5/12) = €26,910.

7. Common Mistakes and Exam Traps

- Capitalising costs incurred after the asset is already capable of operating as management intended, just because it is not yet at full capacity or fully in use — the capitalisation window closes at "capable of operating," not at "actually operating at planned volume."
- Capitalising costs that would have been incurred regardless of the acquisition/construction (e.g. ordinary staffing and utility costs of a store that would run whether or not it had just opened) — only costs necessary to reach the intended operating condition qualify.
- Forgetting that a service contract fee paid alongside an asset purchase is a future-period expense, not part of the asset's cost, even if invoiced together with the purchase.
- Using the specific-borrowing interest rate when the asset was actually funded from a pool of general borrowings — the weighted average rate applies to general borrowings, and mixing the two methods overstates or understates the capitalised amount.
- Forgetting to time-apportion capitalised borrowing costs by the number of months each tranche of expenditure was actually outstanding during the capitalisation period.

8. Must Know Absolutely

Must know:
• PP&E (IAS16 par 6): tangible, used for production/supply/rental/admin, expected use >1 period
• Recognition test: reliably measurable cost **and** probable future economic benefits
• Three cost elements: purchase price + directly attributable costs + estimated dismantling/restoration cost (as a provision)
• Capitalisation stops once the asset is capable of operating as intended — not at full-capacity use
• Excluded from cost: administrative/overhead costs, feasibility studies, staff training, abnormal waste, initial operating losses, relocation/reorganisation costs
• Component approach: depreciate significant parts of an asset separately when their useful lives differ
• Borrowing costs on qualifying assets are capitalised: specific-borrowing rate for dedicated loans, weighted average rate (time-apportioned) for general borrowings

Watch Out

Watch out: the extracted slides for the €100 machine example show a total of €133 built from rows labelled "Invoice and tax (107)", "Transportation (10)", "Calibration (11)" and "Professional fees" without a clean one-to-one correspondence to every individual cost bullet listed on the previous slide (e.g. the €5 storage cost is implicitly excluded rather than explicitly subtracted). The reconstruction above follows the given total and row labels; recompute it yourself from the raw cost list as a self-check.